

Déborah Boyenval

Postdoctoral Fellow
Computational Systems Biology
University of Helsinki, Finland

Phone: +33 642767940

email: deborah.boyenval@helsinki.fi

URL: <https://deborahboyenval.github.io/>

Born: 1994 — Amiens, France

Nationality: French

Current position

2025–present *Postdoctoral fellow*, Systems Biology of Drug Resistance in Cancer (Hautaniemi Lab), University of Helsinki

Academic appointments

2024–2025 Postdoctoral Fellow, Laboratoire Bordelais de Recherche en Informatique (LaBRI), [Synthesis Team](#), Bordeaux, France (Jan. 2024–Jan. 2025).

Formal modeling of age-related changes in hematopoietic cell differentiation from single-cell data, with Loïc Paulevé.

2021–2022 Part-time Teaching and Research Fellow, Computer Science Department, Université Côte d’Azur (Oct. 2021–Sep. 2022).

Areas of specialization

Mathematical modeling (continuous, discrete and stochastic) • Multi-omic analyses • Generative models • Cancer biology

Education

2018–2022 PhD in Computer Science, Université Côte d’Azur, I3S ([SPARKS Team](#)) and Institut de Biologie de Valrose, France (Oct. 2018–Dec. 2022).

Thesis: *Formal modeling of cyclic biological behaviours with checkpoints: the cell cycle regulation*.

Supervisors: Gilles Bernot, Jean-Paul Comet and Franck Delaunay.

2016–2018 MSc in Life Sciences, Université Côte d’Azur, Sophia Antipolis, France.

Major in Bioinformatics and Computational Biology (BIM).

Master’s thesis (2018): *Mathematical modeling of sex-specific chronotoxicity of an anticancer drug: irinotecan*. Supervisor: Annabelle Ballesta (currently at Institut Curie, [Cancer Systems Pharmacology](#)), Paris.

M1 internship (2017): Mathematical modelling of circadian clocks, I3S/iBV.

2013–2016 BSc in Life Sciences, Université Côte d’Azur, Sophia Antipolis, France.

Third-year specialization in Biology, Informatics and Mathematics (BIM).

L3 internship (2016): wet-lab research in the [MINDev Team \(Microbiota, Immunity, and Neurodevelopment\)](#), Institute of Molecular and Cellular Pharmacology, Nice, France.

Grants, honors & awards

2024 PhD Thesis Award, Fondation Université Côte d’Azur.

Publications & posters

PUBLICATIONS

- 2025 Leppiniemi, S., Afenteva, D., Li, Y., Micoli, G., Lavikka, K., Holmström, S., Boyenval, D., Oikkonen, J., Hautaniemi, S., and Muranen, T. A. “SegmentQTL: Identifying genetic variants influencing molecular phenotypes in copy number-driven cancers”. *bioRxiv*. [\[DOI\]](#) [\[PDF\]](#)
- 2024 Chevalier, S., Boyenval, D., Magaña-Lopez, G., Roncalli, T., Vaginay, A., and Paulevé, L. “BoNesis: a Python-based declarative environment for the verification, reprogramming, and synthesis of Most Permissive Boolean networks”. *Computational Methods in Systems Biology (CMSB 2024)*. [\[DOI\]](#)
- 2022 Dulong, S., Souza, L., Machowiak, J., Peuteman, B., Duvallet, G., Boyenval, D., Roth, E., Asgarova, A., Chang, Y., Li, X.-M., Foudi, A., and Ballesta, A. “Sex and Circadian Timing Modulate Oxaliplatin Hematological and Hematopoietic Toxicities”. *Pharmaceutics*, 14, 2465. [\[DOI\]](#)
- 2020 Boyenval, D., Bernot, G., Collavizza, H., and Comet, J.-P. “What is a cell cycle checkpoint? The ToTemBioNet answer”. *Computational Methods in Systems Biology (CMSB 2020)*. [\[DOI\]](#)

DOCTORAL THESIS

- 2022 Boyenval, D. “Formal modeling of cyclic biological behaviours with checkpoints: the cell cycle regulation”. *PhD thesis in Computer Science*.
Université Côte d’Azur; Laboratoire d’Informatique Signaux et Systèmes de Sophia Antipolis (I3S) and Institut de Biologie de Valrose, Nice, France.
Defended December 15, 2022. Supervisors: Gilles Bernot, Jean-Paul Comet, and Franck Delaunay. [\[HAL\]](#)

SUBMITTED MANUSCRIPTS

- 2026 Boyenval, D., Malyutina, A., Oikkonen, J., Liikkanen, S., and Hautaniemi, S. “Synthesis of Ovarian Cancer Multi-Omics Datasets: A Biology-Informed Evaluation Framework”. *Submitted to NeurIPS workshop GenAI4Health and ML4H symposium*.

POSTERS

- 2026 “Generating Synthetic Data from Ovarian Cancer Patients”. RPU Science Day 2026.
- 2021 “Logical and incremental formalization of cell cycle checkpoints”. JOBIM 2021, July.
- 2019 “Introduction of Priorities in Biological Regulatory Networks”. Modelife Annual Meeting, Fréjus, France, November.
- 2019 “A discrete cell cycle model: from phase characterization toward observable properties verification”. Summer School on Formal Modeling of Biological Regulatory Networks, Porquerolles, June.

Teaching

- 2025 Teaching contribution to *Introduction to Systems Biology*, University of Helsinki. Course coordinator: Sampsa Hautaniemi.
- 2018–2021 Teaching Assistant, Université Côte d’Azur (BSc and MSc levels).
Web development, systems and network administration, database theory and SQL, imperative programming and Python, algorithms for biology.

Service & miscellaneous

2019–2021 Treasurer, [ADAMS Doctoral Students' Association](#), Université Côte d'Azur.
2018–2019 FabLab Manager, Valrose campus of Université Côte d'Azur.

Referees

Sampsa Hautaniemi (Postdoctoral referee) — sampsa.hautaniemi@helsinki.fi; +358 29 412 5419.
Jean-Paul Comet (PhD referee) — jean-paul.comet@univ-cotedazur.fr; +33 4 89 15 43 49.
Gilles Bernot (PhD referee) — gilles.bernot@univ-cotedazur.fr; +33 4 89 15 43 84.